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Adolescent pain reports share genetic overlap with adult chronic pain conditions: A polygenic score analysis using the ABCD study

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Abstract

Adolescent pain complaints may be related to genetic risk for chronic pain across the life course. Identifying whether adolescent pain is genetically linked to chronic pain in adulthood can advance understanding of pain etiology and inform early intervention. Two waves of pain assessments were used from the Adolescent Brain Cognitive Development (ABCD) study, a population-based sample of 11,876 adolescents with 94.0% retention across waves. The analyses included 6,387 adolescents of European-like ancestry (mean ages = 12.03 and 12.93 at waves 2 and 3; 52% males). Two polygenic scores (PGSs) were constructed using genome-wide association study summary statistics from up to 435,917 adults in the UK Biobank. One PGS captured shared genetic risk across 24 pain conditions (General Chronic Pain), while the second captured additional musculoskeletal-specific genetic risk across 11 conditions after adjusting for general pain (Musculoskeletal-specific Pain). Mixed-effects models were used to examine associations between these PGSs and adolescent self-reported pain presence, intensity, recurrence, and multi-site pain. Across both waves, 36.0%−37.0% adolescents reported pain. The General Pain PGS was associated with pain presence (b=0.07, OR=1.07, 95%CI=1.02–1.13, FDR-corrected p=0.023) and intensity (b=0.14, 95%CI=0.07–0.21, FDR-corrected p<0.001); but not recurrent pain (b=0.08, OR=1.08, 95%CI=1.01–1.16, FDR-corrected p=0.091) or multi-site pain (b=0.01, OR=1.00, 95%CI=0.94–1.07, FDR-corrected p=0.958). The Musculoskeletal-specific Pain PGS was not significantly associated with the outcomes. Genetic risk for chronic pain in adulthood, as measured by PGSs, is associated with adolescent pain complaints. Adolescent pain signals early vulnerability for chronic pain, highlighting adolescence for early intervention.

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Perspective: This study links adolescent pain to polygenic risk for adult chronic pain, suggesting that early pain reflects enduring genetic liability and reflects central pain processes. These results provide mechanistic insight into chronic pain across the lifespan and highlight adolescence as a period for intervention.

Keywords

chronic pain; adolescent; development; genomic; polygenic score

Introduction

Chronic pain, defined as pain lasting over 3 months, affects approximately 11 to 40% of individuals worldwide[1] and occurs at comparable rates in adolescence[2]. Adolescent chronic pain is associated with worse educational and social outcomes[3], later opioid misuse[4], and increased risk of chronic pain in adulthood[5–8]. Both adolescent and adult pain are heritable[9,10]; however, it remains unclear whether they reflect the same genetic risk.

Adolescent pain frequently persists into adulthood and often spreads across body sites[5–8]. For example, abdominal pain in youth predicts abdominal and multi-site pain 15 years later[8]. Such spreading is characteristic of nociplastic pain, which arises from a sensitized nervous system, rather than tissue damage[1]. Thus, adolescent pain may reflect vulnerability to centralized pain mechanisms that subsequently contribute to chronic pain in adulthood.

Determining whether adolescent pain shares genetic liability with adult pain conditions and whether this liability reflects central versus musculoskeletal genetic influences may clarify adolescent pain etiology. Twin studies indicate that up to 66% of chronic pain variation is attributable to genetic factors that remain stable across adulthood[11,12]. Genomic research shows that many pain conditions partially share genetic architectures, consistent with a cross-condition pain liability[9,13]. Zorina-Lichtenwalter et al.[13] applied genomic structural equation modeling to 24 adult chronic pain conditions and identified two latent genetic factors: a General Chronic Pain factor, underlying all conditions, and an additional Musculoskeletal-specific Pain factor capturing shared genetic variance among 11 conditions beyond the General Pain factor. Biological annotation and correlation analyses of these genetic pain factors revealed a complex interplay of multiple biological systems and processes, including the central nervous system, and genetic associations with cognitive and affective traits[14]. The Musculoskeletal Pain factor was enriched for both central regulatory and peripheral mechanisms (e.g., skeletal development) and shared genetic associations with anthropomorphic traits[13].

The etiological architecture of adolescent pain is less delineated. Adolescent twin studies show mixed results, with some reporting heritability for general chronic pain and specific pain conditions (e.g., migraine)[15–19], and others finding primarily environmental influences for chronic pain conditions across development[17,20–23]. Together, these findings imply continuity of genetic liability from adolescence to adulthood, though direct genomic evidence is lacking.

Polygenic scores (PGSs) can be used to test whether adolescent pain reflects the same genetic influences identified in adult chronic pain. A PGS for a trait is the sum of genetic variants each individual has in their genome, where each variant is weighted by its effect size in a genome-wide association study (GWAS) that has been conducted in an independent sample. A high PGS indicates the individual carries more "risk" variants for the trait. Applying adult chronic pain PGSs to adolescent samples allows us to 1) test whether adolescent pain complaints reflect shared genetic liability with adult chronic pain and 2) provide clues to the shared biological sources, beyond what can be inferred from phenotypic continuity alone. Examining separate PGSs for General Chronic Pain and Musculoskeletal-specific pain can further distinguish whether adolescent pain is linked to broad centralized mechanisms, peripheral musculoskeletal pathways, or both.

Sex differences may further influence adolescent pain. Girls report more chronic pain than boys[24] and such differences often emerge during puberty[25]. Evidence for sex-specific genetic effects is mixed: one twin study found sex-specific genetic effects for recurrent headaches[16], whereas others found none for neck, low back, or general pain[10,19,20]. Thus, whether sex-specific genetic effects contribute to adolescent pain remains uncertain.

The present study explored whether genetic liability for adult chronic pain is associated with adolescent pain in the longitudinal Adolescent Brain Cognitive Development (ABCD) study. Associations between General Chronic Pain and Musculoskeletal-specific Pain PGSs and adolescent self-reported pain presence, intensity, recurrence, and multi-site pain were tested, and potential sex differences were evaluated. This is the first known study to evaluate whether cross-condition chronic pain PGSs are associated with adolescent pain complaints.

Materials and Methods

Study Design and Participants

All behavioral and genetic data were drawn from the Adolescent Brain Cognitive Development (ABCD) study National Data Archive (NDA) data release 5.1. The ABCD study is a United States population-based, longitudinal study following 11,876 youth with yearly assessments beginning at 9–10 years of age. There is annual data collection on behavioral data (e.g., pain reports), as well as neuroimaging, biomarker, and genetic data[26]. The ABCD used sampling techniques to reflect the sociodemographic landscape, including age, gender, race/ethnicity, socioeconomic status, and urbanicity of United States adolescents[27]. The participants were recruited through probability sampling of schools within 21 research sites (information on the sites can be found at: https://abcdstudy.org/consortium_members/).

Parents of the participants provided written informed consent, and the adolescents provided assent prior to participating in the study. Research protocols across the 21 ABCD sites were approved by the University of California–San Diego institutional review board (IRB; protocol number 160091), which is the IRB of record for the entire ABCD Study. The University of California San Diego IRB has stated that analyses using publicly released ABCD data are not human subjects research and do not require their own protocol approval.

An observational cohort study was conducted using yearly follow up 2 (mean age = 12.03, standard deviation (SD) = 0.67, min-max = 10.58–13.83) and follow up 3 (mean age = 12.93, SD = 0.65, min-max = 11.50–14.58) of the ABCD study. Due to the genetically informed nature of this study, adolescents were included if they had behavioral pain data at one of the data collection waves, usable genetic data, and had a genetic ancestry structure consistent with the 1000 Genomes European reference population. The inclusion criteria resulted in 6,387 adolescents (females = 48%; males = 52%; see sample descriptives in Table 1) from the ABCD study. The main analyses only included the European-like ancestry subsample of the ABCD due to a lack of comprehensive, large-scale GWAS of pain data in non-European ancestry samples. This is a pervasive issue in the genetics community which is beginning to be addressed through more inclusive data collection[28]. We examined distributions and analyses of non-European-like ancestries in the ABCD study, shown in Supplementary Tables 1–3. However, as there were not GWASs available for non-European-like ancestries (e.g., African-like, East Asian-like) and PGSs based on GWASs using European-like-ancestry have low portability into other ancestries[29]; the PGS findings in other ancestries should be interpreted with extreme caution.

Measures

Outcomes: Adolescent pain reports: We used self-reported data from the ABCD Pain Questionnaire (version 5.10). Pain presence was measured with the response to “Over the past month, have you had any aches or pains in your body?” (0 = no, 1 = yes). Recurrent pain was defined as reporting pain at both waves, versus only one wave (0 = pain at one wave; 1 = pain at both waves). Pain intensity was assessed using a Numeric Rating Scale (NRS): “Over the past month, how much did it hurt when it was the WORST?” (0 = not at all, 10 = the worst). The NRS is a valid self-report pain intensity measure for adolescents[30]. The Pain Questionnaire does not include a measure of pain chronicity (pain lasting 3+ months).

Pain location(s) was assessed using the pediatric adaptation of the Collaborative Health Outcomes Information Registry (Peds-CHOIR) body location map[31]. Up to 74 locations could be selected for males and females. Body maps are valid and reliable instruments for pain location in adolescents[32]. Multisite pain was coded using criteria from previous work: “pain in 2+ major body regions (i.e., head/neck/face, abdomen/pelvic, or musculoskeletal), non-contiguous pain, or whole-body pain (i.e., pain identified across two or more of the following distinct regions: head/neck, abdomen/pelvis, chest, back, right upper extremity, left upper extremity, right lower extremity, left lower extremity)”[33(p3)].

Exposure: Polygenic Scores: The primary exposures were PGSs for (1) A General Chronic Pain factor, which captured the shared genetic variance that contribute to 24 pain conditions, and (2) A Musculoskeletal-specific Pain factor, which captured genetic variance specific to 11 musculoskeletal pain conditions[13]. PGSs are calculated by using the effect sizes of alleles in a discovery GWAS to “score” the genome of each individual in an independent target sample; each individual’s PGS is essentially the sum of all their alleles weighted by the discovery GWAS’s effect sizes[34]. Thus, it provides an efficient summary of their genetic liability for the trait of interest.

The discovery GWASs were conducted in a previous study[13] using the UK Biobank (UKBB), a United Kingdom-based study with deep phenotyping and genotyping data collected on about 500,000 individuals, aged 40–69 years[35,36]. Up to 435,917 UKBB participants who were recruited between 2006 and 2010 (54% female; 46% male) were included in the pain GWASs. Zorina-Lichtenwalter et al.[13] included 24 pain phenotypes: headache, migraine, neck/shoulder pain, back pain, chest pain/discomfort, chest pain during physical activity, irritable bowel syndrome, gastritis, esophagitis, stomach pain, carpal tunnel, cystitis, hip pain, knee pain, leg pain, gout, enthesopathies of lower limb, hip arthrosis, knee arthrosis, enthesopathies, rheumatoid arthritis, arthropathies nonspecific including osteoarthritis, pain in join, and chronic widespread pain. Chronic pain was defined as pain lasting 3 months at a location (e.g., chronic knee pain). Conditions with persistent pain as a major symptom (e.g., osteoarthritis) were also included. They used GWASs of these 24 conditions to estimate a genomic structural equation model[37], then conducted multivariate GWASs (effective N= 422,752) on the two latent factors in their model (shown in Figure 1). GWAS analyses were based on individuals of European-like ancestry to minimize population stratification and due to a lack of well-powered, pain data available for non-European-like ancestries. Although the discovery GWASs were based on those of European-like ancestry in the United Kingdom, the PGSs derived from this source will have transferability to European-like ancestry populations in the United States. The need for transferability across genetically European-like populations was a major reason present analyses were restricted to participants of European-like ancestry[38].

Genotyping and Polygenic Score Construction: Figure 2 visualizes the process of deriving and constructing polygenic scores. Genotyping of saliva samples of the ABCD study was performed using the Smokescreen Array at baseline[39]. Genotypes were imputed using the Haplotype Reference Consortium reference panel[40]. After imputation, data were filtered based on imputation quality (<0.80), minor allele frequency (<0.01), sample missingness (>0.05), and Hardy-Weinberg equilibrium (p<1e-10). Genetic ancestry principal components (PCs) were then derived by projecting ABCD genomes onto PCs from the 1000 Genomes Project reference population[41]. Individuals were classified as having European-similar genetic ancestry if their PC scores were within 5 SD from the mean of the European reference on all of the first four principal components. It is important to note that ancestry is not synonymous with race or ethnicity. An individual's genetic ancestry may project onto an ancestry that is different from what the individual primarily identifies as ethnically and/or racially.

The PGSs were generated using polygenic risk score – continuous shrinkage (PRS-CS)[42].

1. The procedure of PRS-CS is as follows:
2. PRS-CS was provided summary association statistics from a GWAS conducted in the discovery sample (here UKBB) and the genotypes of the target sample (here ABCD).
3. PRS-CS used a Bayesian method that places continuous shrinkage priors on SNP effect sizes. Posterior SNP effect sizes were estimated controlling for

the non-independence of genetic variants, quantified with a reference linkage disequilibrium panel.

4. Individual-level PGSs in the ABCD target sample were calculated using PLINK's --score function, which sums weighted effects of genetic variants to produce a PGS for each participant[43].

5. The PGSs were standardized prior to analysis.

Covariates: All models included biological sex and the first 10 genetic PCs as covariates. Genetic PCs control for residual population structure, minimizing confounding due to allele frequency differences correlating with non-genetic factors such as geographic variation or socioeconomic status. We did not adjust for other psychosocial risk factors of chronic pain as these variables may be correlated with the child's polygenic score due to shared risk genetic factors (e.g., genetic risk influencing both depression and pain) or through gene-environment correlations (e.g., parental genotype influencing the home environment)[44,45]. Conditioning on such variables could attenuate the association of interest and answers different hypotheses, such as, does adult-derived genetic risk for chronic pain share a genetic association with adolescent pain above and beyond shared genetic risk with depression?

Statistical Analysis

The analyses were performed in R version 4.5.0 using the packages CHOIRBM[46,47] and lme4[48]. We visualized endorsed pain locations by biological sex and wave using the CHOIRBM functions plot_male_choirbm() and plot_female_choirbm(). Multilevel logistic regression models were fit for binary outcomes (pain presence, recurrent pain, and multisite pain), and multilevel linear regression for the continuous pain intensity outcome. Predictors were PGSs and covariates. All models included random intercepts for family ID to control for shared familial environmental effects, and random intercepts for subject ID to adjust for within-person clustering due to repeated measures.

The pain presence, pain intensity, and multisite pain models used pain variables measured at both wave 2 and wave 3, with a random intercept for subject to account for the non-independence of the repeated measures. The recurrent pain model used a variable that was derived across both timepoints, so did not include the within-person random intercept. For pain presence, the control group reported no pain, enabling us to test whether the pain PGSs can differentiate between adolescents with and without pain. For recurrent pain, the control group reported pain at a single timepoint, enabling us to test whether the PGSs can differentiate between adolescents with transient and recurrent pain. The multi-site pain model included multi-site pain measured at both timepoints.

For multi-site pain, the control group reported pain at a single-site, and multi-site pain was defined as reporting pain at 2+ non-congruent body locations. This comparison enabled us to test whether the PGS is associated with having pain at more than one location. Sensitivity analyses were conducted to evaluate alternative operationalizations of multi-site pain. First, we created an "ever multi-site pain" phenotype, coded as 1 if adolescents reported multi-site pain at either wave or 0 if they reported single-site pain at both waves. Second, we created a "spreading" phenotype, coded as 1 for adolescents who reported single-site pain at Wave 2 and multi-site pain at Wave 3 and 0 for those who reported single-site pain at both waves.

A significant association between the PGS and pain outcome suggests that higher polygenic liability for chronic or musculoskeletal pain reported in older adulthood is associated with a greater probability of reporting a pain experience in adolescence. To test whether associations between PGSs and pain outcomes varied by sex, we added an interaction term between each PGS and biological sex (coded 0=male, 1=female) to all the primary models. A significant interaction term coefficient would indicate that the effect of the PGS depends on biological sex.

The two waves of pain data used in the study had a high retention rate (94%). All available data were used, including people who were missing pain data at a single timepoint, rather than using list-wise deletion. People who were missing pain data at timepoint 2 did not differ on pain at timepoint 1 ($p>0.05$), suggesting that data were "missing at random." Multiple testing was controlled using the false discovery rate (FDR) procedure for eight tests with statistical significance set at FDR-corrected $p<0.05$[49].

Results

Descriptive statistics are provided in Table 1. Pain location endorsements by sex and wave are provided in Table 2 and visualized in Figure 3. At mean age 12, 36% of the sample reported experiencing pain, and 62% of those experiencing pain reported pain in multiple locations. At mean age 13, 37% reported pain, and 63% of those experiencing pain reported pain in multiple locations. Among participants who completed the pain questionnaire at both time points (n=3,917), 30.2% reported pain at both waves. Musculoskeletal pain was the most endorsed pain type overall and was more frequently reported by males. Females endorsed more abdominal/pelvic pain than did males. There was no significant sex difference in headache endorsement.

The PGS regression results are provided in Table 3. The General Pain PGS was significantly associated with higher odds of pain presence (odds ratio (OR) = 1.07, 95% confidence intervals (95% CI) [1.02, 1.13], FDR-corrected p = 0.023). The General Pain PGS was also associated with greater pain intensity in those with pain (b = 0.14, standard error (SE) = 0.04, FDR-corrected p = 0.001). After FDR-correction, the General Pain factor PGS did not significantly associate with recurrent pain (OR = 1.08, 95% CI [1.01, 1.16], FDR-corrected p = 0.097) or multi-site pain (OR = 1.01, 95% CI [0.94, 1.07], FDR-corrected p = 0.958).

The Musculoskeletal-specific Pain PGS was not significantly associated with pain presence (OR = 1.04, 95% CI [0.99, 1.09], FDR-corrected p = 0.282), pain intensity (b = 0.00, SE = 0.04, FDR-corrected p = 0.958), recurrent pain (OR = 1.01, 95% CI [0.95, 1.09], FDR-corrected p = 0.958), nor multi-site pain (OR = 1.01, 95% CI [0.94, 1.08], FDR-corrected p = 0.958).

As sensitivity analyses, we examined two alternative multi-site pain phenotypes: (1) "ever multi-site pain" (n = 188 single-site at both timepoints; n = 2,261 multi-site at either timepoint) and (2) "spreading" from wave 2 single-site to wave 3 multi-site (n = 188 no spreading, $n = 228$ spreading). The ever multi-site pain phenotype was not significantly associated with the General Pain PGS (OR = 1.01, 95% CI [0.88, 1.18], uncorrected $p = 0.892$) or the Musculoskeletal Pain PGS (OR = 0.97, 95% CI [0.83, 1.13], uncorrected $p = 0.662$). Similarly, the spreading pain phenotype also did not significantly associate with the General Pain PGS (OR = 1.15, 95% CI [0.94, 1.40], uncorrected $p = 0.184$) nor the Musculoskeletal Pain PGS (OR = 0.89, 95% CI [0.72, 1.10], uncorrected $p = 0.274$). Although effect estimates for spreading pain were larger, none reached statistical significance, likely due to the smaller sample size.

In models that included PGS × Sex interaction terms, the main effect of the General Pain PGS on pain outcomes remained significant. However, none of the PGS × Sex interaction terms were significant for either pain factor (e.g., uncorrected $p > 0.05$ for all models).

Discussion

Within a population-based cohort of United States adolescents, over 30% of adolescents reported pain, and 20% reported non-contiguous, multi-site pain. Pain in musculoskeletal regions was most endorsed, especially among males, while females more often reported abdominal/pelvic pain. Polygenic risk for a cross-condition General Chronic Pain factor was significantly related to adolescent pain presence and intensity. Although musculoskeletal regions were the most frequently endorsed, only the General Chronic Pain PGSs, and not the PGSs for pain specific to Musculoskeletal Pain, were significantly associated with adolescent pain reports. These findings suggest that adolescent pain may partly reflect genetic liability to general, centrally mediated chronic pain throughout adulthood, and are genetically distinct from musculoskeletal-specific pain in adulthood.

The General Chronic Pain PGS was associated with both pain presence and intensity in early adolescence, suggesting that the etiological sources of adolescent pain may reflect emerging vulnerability to centralized pain mechanisms. This finding is notable because the General Chronic Pain PGS had many indicators for pain conditions not usually diagnosed in children (e.g., osteoarthritis)[13]. Prior research found that a rheumatoid arthritis PGS was associated with cognitive phenotypes in children[50], and that chronic multi-site and chronic back pain PGSs were related to child psychopathology in an earlier wave of ABCD data (mean age = 9.92)[51]. These findings are consistent with the current results, as the multivariate adult pain PGS, enriched for central nervous system pathways and affective traits, is linked to adolescent pain. Future research may examine how psychopathology PGSs, such as anxiety and depression, mediate the association of the pain PGS with adolescent outcomes.

The enrichment of the central nervous system for the General Pain factor[13] on which the pain PGS was based supports the hypothesis that adolescent pain may reflect early manifestations of centralized pain. Prospective and retrospective studies have found that pain in adolescence increases risk of spreading pain in adulthood; a hallmark feature of nociplastic pain. The early pain pathway hypothesis suggests that adulthood chronic pain may be caused by adolescent pain that incites maladaptive neuronal changes at a developmentally sensitive time[52]. Experiencing pain in infancy leads to maladaptive responses in pain processing pathways in animal models[52]. In healthy humans, repetitive pain stimuli lead to an activation of inhibitory projections from the amygdala down through the dorsal horn neurons that inhibit pain. These pathways are attenuated in neonates who have experienced pain: The functional integrity of the descending inhibitory systems worsened and was shown to lead to an over-processing of pain stimuli, or chronified pain[52]. Early-life pain has been associated with poorer descending inhibition of pain in humans[53].

While these studies suggest a causal component, the genetic associations examined here do not speak to causality: This maladaptation of descending control mechanisms may be direct, due to shared genetic risk, or a mix. The findings suggest that adolescent pain relates to adulthood risk of centrally augmented pain at least partially through an increased genetic propensity at the individual level. Clinically, adolescence may be a salient intervention point to prevent exacerbated pain throughout adulthood. In one long-term follow up study, 72% of youth patients who underwent intensive interdisciplinary pain treatment showed remission or reduction in chronic pain grade severity seven years after treatment; however, 41.6% continued to experience chronic pain, and former patients reported worse physical and mental health status and greater health care utilization compared to controls[54]. While PGSs do not generally perform well for individual-level prediction of chronic pain incidence[55], future research may assess the utility of pain PGSs in predicting response to certain clinical interventions (e.g., cognitive-behavioral therapy).

There was no significant relationship between the General Pain PGS and multisite pain. This lack of association may seem counterintuitive, but prior research has shown that in adolescence, even single-site pain may represent a unified latent construct[56], and that additional pain locations often emerge sequentially, typically within a year after the first pain onset[57]. The follow-up period may not have been long enough to detect the emergence or spread of multisite pain among genetically at-risk youth.

Consistent with epidemiological data, musculoskeletal pain was frequently endorsed[2], yet the Musculoskeletal-specific Pain PGS was not associated with any adolescent pain outcomes. This factor included genetic liability related to skeletal development and peripheral mechanisms, which may be less salient during adolescence and more related to aging processes later in adulthood. While genetic influences on pain remain stable across adulthood[11], genetic influences were not found in twin models of the age 12 pain reports, and only began to emerge a year later[22]. Hence, pain risk represented by the Musculoskeletal-specific Pain PGS may emerge later in life when structural and degenerative processes become more influential.

Significant sex differences emerged in the endorsement of pain across body regions: females reported more abdominal/pelvic pain than males, whereas males reported more musculoskeletal pain. No significant differences were observed for headaches. A systematic review of adolescent pain found that females generally report more pain than males; females tended to report more headaches, abdominal, and multisite pain than males, and there were no significant differences in musculoskeletal pain[24], a pattern that partially diverges from the current findings.

Despite these prevalence differences, there was no evidence for sex-specific associations between the pain PGSs and adolescent pain outcomes. This pattern aligns with twin research on adolescent neck pain showing similar magnitude of heritability estimates and sources of heritability across sex[19]. Although a genomic study indicated partial sex differences in the adult genetic architecture of multisite chronic pain, genetic effects were largely shared among males and females[58]. The absence of genetic sex effects does contradict some twin studies, which have found different genetic etiologies for girls and boys[16,18–20].

Although the findings suggest overlap in these genetic effects across biological sex, they do not rule out the existence of sex-specific genetic effects on adolescent pain. Future GWASs of adolescent pain and subsequent sex-stratified analyses may reveal that genomic architecture differs in boys and girls, but that differential genetic effects may be developmentally specific[25]. The pain PGSs were derived from an adult sample and would not capture these developmentally specific variations. Furthermore, sex effects may be body site- or pain condition-specific: for example, headaches and migraines may have unique etiological pathways that differ by sex and may be differentially affected by puberty[16,59]. While the genetic effects examined here were not distinguished by sex, future work could assess whether sex modifies genetic risk for specific pain conditions.

This work demonstrates that genomic findings from adult pain are relevant to child pain, extending theoretical insights from adult studies to younger developmental periods. Identifying shared genetic liability for chronic pain across development refines mechanistic understanding and highlights adolescence as a key period for prevention. Although current PGSs lack clinical utility and perform poorly for individual-level risk prediction[55], integrating genetic information into developmental models of pain can advance understanding of what distinguishes transient pain from persisting, centralized pain. Furthermore, PGSs can be used to enhance mechanistic understanding: Polygenic risk for General Chronic Pain, enriched for central nervous system pathways and overlapping with affective/cognitive traits, was associated with adolescent pain. In contrast, the Musculoskeletal-specific Pain PGS, indexed by skeletal processes and overlapping with anthropomorphic traits, was not associated with adolescent pain outcomes. This distinction suggests that early pain reports may be driven more by central sensitization and affective–cognitive mechanisms than by peripheral musculoskeletal processes.

The demonstrated relevance of adults' PGSs for adolescent pain also opens the door to application of other models to the data. PGSs can also be used to test gene-environment interactions and potentially improve risk stratification. For example, youth at higher genetic risk for pain may be more sensitive to environmental risk factors for pain (e.g., stressors or injuries) that ultimately trigger pain chronification. Further, gene-environment correlations can also be tested by examining whether pain-related PGSs are correlated with environmental risk factors (e.g., parenting style). Future work could formally evaluate such hypotheses.

The major strength of this study is the use of a well-powered, multivariate General Chronic Pain GWAS of adults applied to a large, prospective adolescent cohort. However, several limitations should be considered. First, the ABCD study lacks measures of pain chronicity (i.e., duration >3 months) making it difficult to distinguish between transient growing, acute, and chronic pain complaints. Second, the recurrent pain measure does not differentiate between persistent and intermittent pain. Third, the effect sizes are not large, but this is consistent with expected effect sizes from other PGS studies. As GWASs of pain reach even larger sample sizes, the strength of these genetic associations will be estimated more precisely. Lastly, the analyses were limited to participants of European-like ancestry due to the lack of well-powered genetic datasets of non-European-like ancestry (e.g., African-like, East Asian-like), limiting generalizability.

In conclusion, these findings improve mechanistic understanding of adolescent pain. Adolescent pain complaints may signal early biological vulnerability to chronic pain later in life. Recognition of adolescent pain complaints as a meaningful clinical indicator highlights this age as a critical window for intervention, with the potential to alter long-term chronic pain trajectories, and improve multiple outcomes across the lifespan.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Disclosures

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Data Statement

All data used in this publication are open access and available to bona fide researchers through processes detailed at https://www.ukbiobank.ac.uk/ and https://abcdstudy.org/ for United Kingdom Biobank and ABCD, respectively. Researchers who have been approved by the National Institute of Mental Health Data Archive (NDA) Data Use Certification may access the ABCD study data (https://nda.nih.gov/abcd/). The ABCD data repository grows and changes over time. The ABCD data used in this report came from the ABCD release 5.1.

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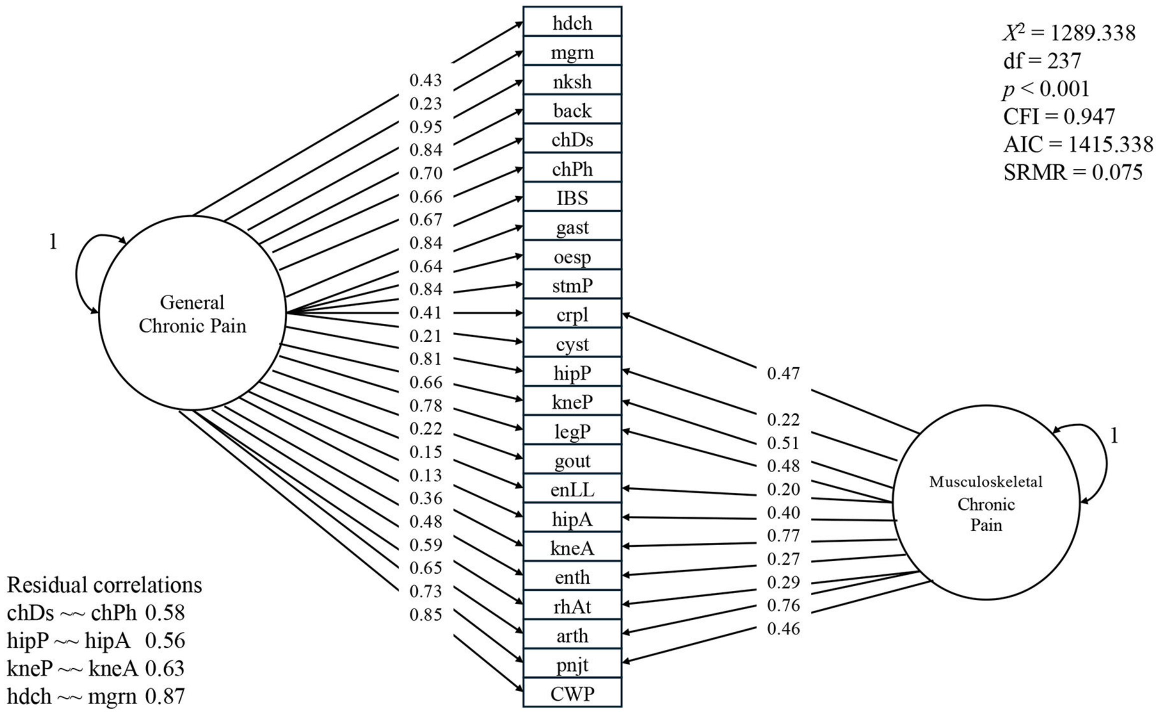

Figure 1.

A visualization of the bifactor-like genetic model of chronic pain.

A bifactor-like genetic model of chronic pain was fit based on past literature (12). Each rectangle represents GWAS summary statistics of a pain condition. The General Chronic Pain circle represents a latent factor that is capturing the covariance of all 24 pain indicators. The Musculoskeletal Chronic Pain circle represents a latent factor that is capturing shared variance of 11 pain conditions. The numbers represent the strength of the loading onto the factor, with 0 being no shared covariance and 1.00 being full shared variance with the factor. The factors' variances were standardized to 1. Residual correlations between some pain conditions were included. The residuals of each individual pain condition are not visualized, but can be found elsewhere (12).

hdch = headache; mgrn = migraine; nksh = neck/shoulder; back = back; chDs = chest pain/discomfort; chPh = chest pain during physical activity; IBS = irritable bowel syndrome; gast = gastritis; oesp = oesophagitis; stmP = stomach pain; crpl = carpal tunnel; cyst = cystitis; hipP = hip pain; kneP = knee pain; legP = leg pain; gout = gout; enLL = enthesopathies of lower limb; hipA = hip arthrosis; kneA = knee arthrosis; enth = enthesopathies; otRA = rheumatoid arthritis; arth = arthropathies nonspecific including osteoarthritis; pnjt = pain in joint; CWP = chronic widespread pain.

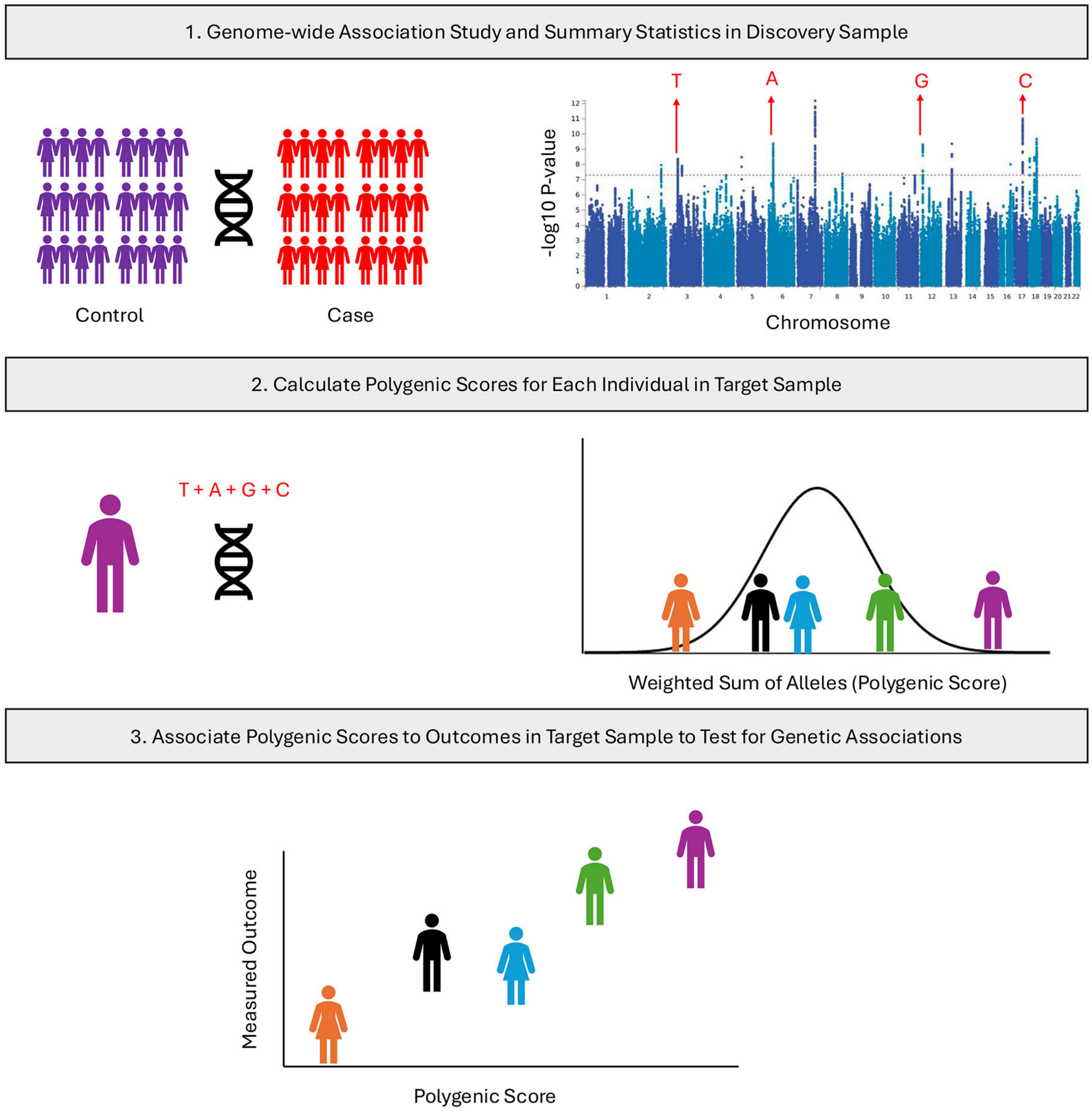

Figure 2.

The process of constructing polygenic scores.

A Genome-Wide Association Study (GWAS) is conducted in a discovery sample to identify genetic risk variants associated with the trait of interest. The trait of interest can either be a case-control or continuous phenotype. The summary statistics from this GWAS are used to constructed weighted sums or alleles, or polygenic scores, in a target sample. The polygenic scores can then be associated to measured outcomes in the target sample to test for genetic associations.

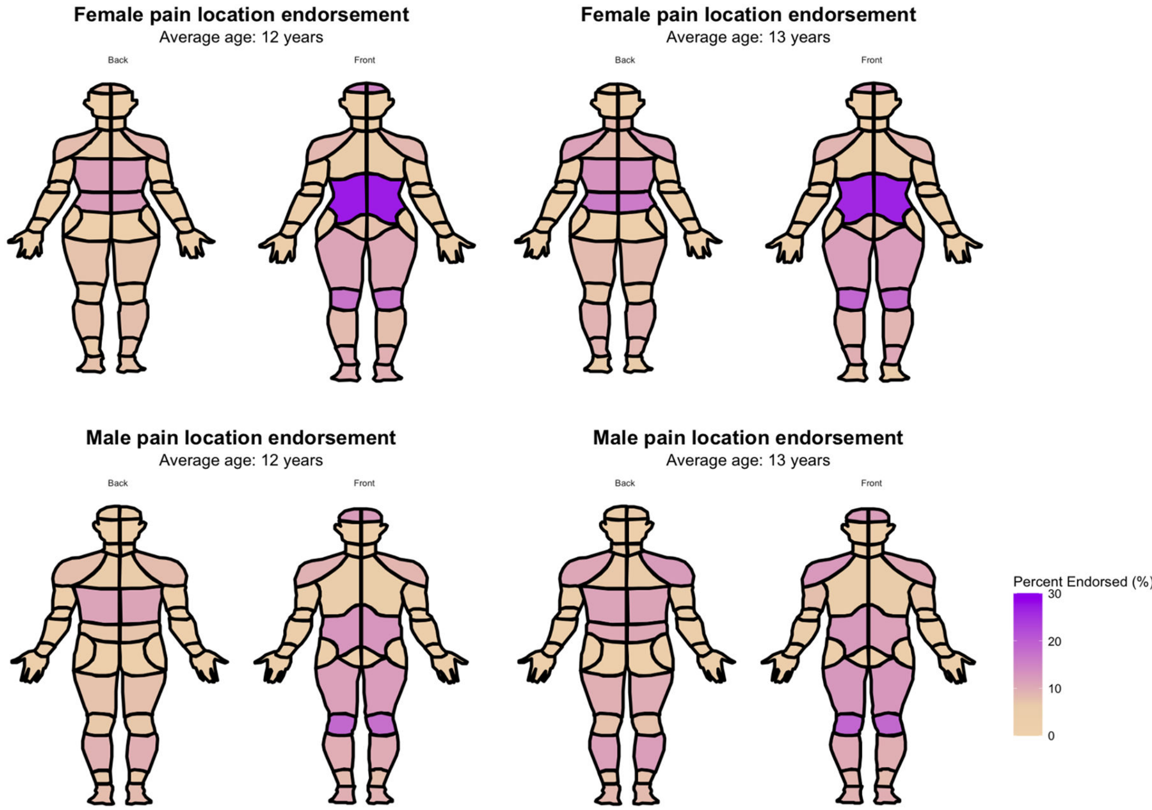

Figure 3.

Endorsed pain locations of adolescents at two timepoints split by sex

The body maps visualize the locations of pain endorsement of adolescents who reported pain.

Location(s) of pain was assessed using the pediatric adaptation of the Collaborative Health Outcomes Information Registry (Peds-CHOIR) body location map (21), which includes 74 locations that could be selected for females and males. At age 12, 1,804 females and 1,188 males report pain. At age 13, 1,118 females and 1,117 males report pain. The plots were generated using the R Package “CHOIRBM” (29–30).

Table 1

Descriptive statistics of pain phenotypes in the ABCD study

Measures	***n***	**Mean (SD) or %**	**Min-Max**
Biological Sex	6,387		-
Males		53.2%	
Females		46.8%	
Gender	6,387		-
Boys		53.2%	
Girls		46.7%	
Gender queer/non-conforming		<1%	
Different identity		<1%	
Don’t know		<1%	
Ethnicity	6,381		-
Hispanic/Latinx		7.8%	
Non-Hispanic/Latinx		91%	
Don’t know		<1%	
Race	6,379		-
American Indian/Alaska Native		2.6%	
Asian		<1%	
Black or African American		1.3%	
Don’t know		<1%	
Native Hawaiian/Pacific Islander		<1%	
Other Race		1.4%	
White		93.8%	
Recurrent pain	2,160		-
Yes recurrent pain		18.5%	
No recurrent pain		15.3%	
First follow up			
Age	6,295	12.03 (0.67) years	10.58–13.83
Pain last month	6,279		
Yes pain		36.3%	
No pain		63.7%	

Measures	*n*	Mean (SD) or %	Min-Max
Pain intensity	2,282	5.93 (2.28)	0–10
Multi-site pain	2,272		-
Multi-site pain		21.9%	
Single-site pain		13.7%	
Second follow up			
Age	6,041	12.93 (0.65) years	11.50–14.58
Pain last month	6,011		-
Yes pain		37.3%	
No pain		62.7%	
Pain intensity	2,244	5.93 (2.25)	0–10
Multi-site pain	2,235		-
Multi-site pain		22.2%	
Single-site pain		12.8%	

Note. These descriptive statistics are on pain outcomes of 6,387 adolescents of European-like ancestry from the ABCD study. Means with standard deviations (SD) and percentages (%) are presented for continuous and categorical variables, respectively. Descriptive statistics are presented for two waves of variables that are used in the study. Pain intensity is described for all those who reported a pain episode in the preceding month. Descriptive statistics for recurrent pain and multi-site pain are out of the total sample including those without pain. Dashes indicate not applicable for categorical data.

Table 2

Descriptives of pain and pain locations by sex and wave

Measures	Pain Total	Pain Females	Pain Males	Sex difference *p*
Recurrent pain	54.8%	58.47%	51.54%	**$p = 0.001$**
Age 12				
Pain presence[t]	36.3%	37.0%	35.7%	$p = 0.290$
Pain intensity	5.93 (2.28)	6.08 (2.25)	5.79 (2.30)	**$p = 0.003$**
Multisite pain	61.5%	63.2%	60.0%	$p = 0.120$
Headache	23.6%	25.1%	22.2%	$p = 0.108$
Abdomen/pelvic	21.5%	29.9%	13.7%	**$p < 0.001$**
Musculoskeletal: Extremities/back/chest	82.0%	79.2%	84.5%	**$p = 0.001$**
Age 13				
Pain presence[t]	37.3%	40.0%	35.0%	**$p < 0.001$**
Pain intensity	5.93 (2.25)	6.21 (2.15)	5.65 (2.3)	**$p < 0.001$**
Multisite pain	63.5%	63.3%	63.7%	$p = 0.873$
Headache	22.1%	22.9%	21.3%	$p = 0.367$
Abdomen/pelvic	21.6%	30.2%	13.0%	**$p < 0.001$**
Musculoskeletal: Extremities/back/chest	84.0%	80.0%	88.1%	**$p < 0.001$**

Note. The count and frequencies of endorsed pain qualities and locations in people who report pain by year and sex. For pain intensity, the mean is reported. Descriptively, two-proportion z-tests were conducted to assess biological sex differences in pain locations, multisite pain, and recurrent pain. Two-tailed t-tests were conducted to assess biological sex differences in mean (standard deviation) pain intensity. The extremities/back/chest is comprised of 6 musculoskeletal regions: The back (low back, middle back, and upper back), chest, arms (hand, wrist, forearm, elbow, bicep, shoulder), and legs (foot, ankle, calf, knee, and thigh). Bolded font signifies significance at a $p<0.05$ level.

[t] Pain presence percentages were calculated from the whole sample

Table 3

Associations between adolescent pain reports and pain polygenic scores

Pain outcomes	Model *n* (unique individuals *n*)	Predictor	Estimate	Odds Ratio	95% CI	*p*	FDR-corrected *p*
Pain presence	12,290 (6,387)	Pain PGS	**0.07** [0.03]	**1.07**	**[1.02, 1.13]**	**0.006**	**0.023**
		MSK PGS	0.04 [0.03]	1.04	[0.99, 1.09]	0.141	0.282
		Sex	**0.17** [0.05]	**1.18**	**[1.07, 1.31]**	**0.001**	-
Pain intensity	4,526 (3,342)	Pain PGS	**0.14** [0.04]	-	**[0.07, 0.21]**	**<0.001**	**0.001**
		MSK PGS	0.00 [0.04]	-	[−0.08, 0.07]	0.958	0.958
		Sex	**0.41** [0.07]	-	**[0.26, 0.55]**	**<0.001**	-
Recurrent pain	5,903 (5,903)	Pain PGS	**0.08 [0.04]**	**1.08**	**[1.01, 1.16]**	**0.034**	0.091
		MSK PGS	0.01 [0.04]	1.01	[0.95, 1.09]	0.681	0.958
		Sex	**0.20 [0.07]**	**1.22**	**[1.06, 1.40]**	**0.005**	-
Multi-site pain	4,507 (3,331)	Pain PGS	0.00 [0.03]	1.00	[0.94, 1.07]	0.907	0.958
		MSK PGS	0.01 [0.04]	1.01	[0.94, 1.08]	0.807	0.958
		Sex	0.06 [0.07]	1.07	[0.93, 1.22]	0.355	-

Note. The estimates displayed are the beta and log odds changes for the outcome variables for every 1 standard deviation increase in the PGSs. The significant estimates indicate that on average, as genetic risk for adult general pain increases for an individual, their risk of the pain complaints significantly increases. The sample size for each of these models varies based on who was included in the control group. Logistic regression with a logit link function were conducted for pain presence, recurrent pain, and multi-site pain as these outcome variables were categorical. The control group for pain presence was reporting no pain at either timepoint. The control group for recurrent pain was reporting pain at only one timepoint. The control group for multi-site pain was reporting single-site pain. Linear regression was conducted for pain intensity as this outcome variable was continuous; only those who reported having pain were included in this analysis. The pain presence, intensity, and multisite pain models included random intercepts for subject ID, to control for the two waves of data, and for family ID, to control for relatedness in the sample. The model *n* represents the number of pain observations included. The unique individual *n* represents the number of individual subjects, some who had more than one wave of pain observations. The recurrent pain model did not include a random intercept for subject ID because there was only one observation per subject based on both time points. Biological sex and 10 ancestry principal components were controlled for in every model (ancestry PC effects are not listed). Males are coded as 0 and females as 1 for sex. Bold font indicates significance. MSK= Musculoskeletal-specific Pain factor; 95% CI = 95% confidence intervals.